

Prison Management System

Key Entities

The entities derived from the functional requirements (FR-001 to FR-028) and represent the major people, records, resources and processes the system manages.

1. Entity Overview

#	Entity	Group	Requirements
1	User	Access and Administration	FR-001 to FR-004
2	Role	Access and Administration	FR-002, FR-004
3	Inmate	Inmate and Custody	FR-005, FR-006
4	Cell	Inmate and Custody	FR-007, FR-008
5	CellAssignment	Inmate and Custody	FR-007, FR-009
6	Case	Legal and Sentencing	FR-013 to FR-015
7	Crime	Legal and Sentencing	FR-013
8	Sentence	Legal and Sentencing	FR-019 to FR-021
9	Incident	Discipline	FR-010 to FR-012
10	Visitor	Visitors	FR-016
11	Visit	Visitors	FR-016 to FR-018
12	RehabProgram	Rehabilitation	FR-022
13	Enrollment	Rehabilitation	FR-022, FR-024
14	AttendanceRecord	Rehabilitation	FR-023
15	AuditLog	System Support	FR-025, FR-026
16	Notification	System Support	FR-027, FR-028

2. Entity Definitions

Access and Administration

1. User

Requirements: FR-001 to FR-004 **Purpose:**

A staff account.

Attributes: user_id, full_name, username, password_hash, email, role_id, status (active/deactivated), created_at

Relationships:

- Belongs to one Role.
- Performs many Incidents, Visits and AuditLog entries.

- Receives many Notifications.

2. Role

Requirements: FR-002, FR-004

Purpose: Defines permissions (Super Admin, Records, Disciplinary, Medical officer, Case, Visitor and Rehabilitation Officer).

Attributes: role_id, role_name, permissions

Relationships:

- One Role has many Users.

Inmate and Custody

3. Inmate

Requirements: FR-005, FR-006

Purpose: The central entity of the system.

Attributes: inmate_id (unique), full_name, gender, date_of_birth, national_id, admission_date, photo, next_of_kin, classification, status

Relationships:

- Has many Cases, Incidents (via incidentInmate), Visits, Enrollments and CellAssignments.

4. Cell

Requirements: FR-007, FR-008

Purpose: A housing unit.

Attributes: cell_id, cell_number, block, capacity, current_occupancy, classification_type

Relationships:

- Has many CellAssignments.

5. CellAssignment

Requirements: FR-007, FR-009

Purpose: Resolves the Inmate-Cell many-to-many relationship and keeps transfer history.

Attributes: assignment_id, inmate_id, cell_id, date_assigned, date_released, assigned_by

Relationships:

- Belongs to one Inmate and one Cell.

Health management

6. Health records

Requirements: FR-29, FR-34

Purpose: The health profile of an inmate created at enrollment together with initial screening monitored by health officer

Attributes: Health_record_id, inmate_id, screening_date, screening_results, blood_group, created_by, created_at, status (ongoing/critical/recovered)

Relationships:

- Belongs to one inmate
- Has many MedicalConditions, Consultations, and MedicalAppointments
- Created by one User (medical officer)

7. MedicalCondition

Requirements: FR-30

Purpose: To have a record of a known medical condition/allergy of an inmate

Attributes: condition_id, health_record_id, condition_type (condition/allergy), condition_name, severity, date_recorded, status(active/resolved), recorded_by

Relationships:

- Belongs to one HealthRecord
- Recorded by one User

8. Consultation

Requirements: FR-31

Purpose: A medical consultation, including the diagnosis and treatment provided

Attributes: consultation_id, health_record_id, consultation_date, complaint, diagnosis, treatment_provided, consulted_by

Relationships:

- Belong to one HealthRecord
- Has many Medications prescribed during consultation
- Conducted by one User

9. Medication

Requirements: FR-032

Purpose: to record medications prescribed during/after consultation.

Attributes: medication_id, consultation_id, drug_name, dosage, frequency, start_date, end_date, status, prescribed_by

Relationships:

- Belongs to one consultation
- Has many MedicationAdministrations
- Prescribed by one User

10. MedicationAdministration

Requirements: FR-32

Purpose: A record of each time a prescribed medication is given to the inmate

Attributes: administration_id, medication_id, administrated_datetime, dose_given, administered_by

Relationships:

- Belongs to one Medication
- Recorded by one user

11. MedicalAppointment

Requirements: FR-033, FR-36

Purpose: A scheduled medical appointment or follow-up, including referrals to external facilities

Attributes: appointment_id, health_record_id, appointment_datetime, purpose, appointment_type (internal/external), external_facility, status (scheduled/completed/cancelled) scheduled_by

Relationships:

- Belongs to one HealthRecord
- Scheduled by one User
- Referenced by Notifications for reminders

Legal and Sentencing

12. Case

Requirements: FR-013 to FR-015

Purpose: The legal case.

Attributes: case_id, inmate_id, case_number, court, case_start_date, case_end_date, status (ongoing/closed/appealed), recorded_by

Relationships:

- Belongs to one Inmate.
- Has many Crimes (through CaseCrime) and one or more Sentences.

13. Crime

Requirements: FR-013

Purpose: A reference list of offences.

Attributes: crime_id, offence_name, description, category

Relationships:

- Many-to-many with Case through CaseCrime (case_id, crime_id).

14. CaseCrime

Requirements: FR-013

Purpose: links a case to the offence/crime. Resolves the many to many relation between case and crime

Attributes: case_id, crime_id (composite primary key)

15. Sentence

Requirements: FR-019 to FR-021

Purpose: The penalty attached to a case.

Attributes: sentence_id, case_id, inmate_id, duration, start_date, release_date, status (active/completed/commuted)

Relationships:

- Belongs to one Case and one Inmate.
- Triggers release Notifications.

Discipline**16. Incident**

Requirements: FR-010 to FR-012 **Purpose:**

A disciplinary event.

Attributes: incident_id, incident_date, description, status (open/resolved), outcome, reported_by

Relationships:

- Reported by one User.
- Involves many Inmates through IncidentInmate (incident_id, inmate_id, role_in_incident).

17. IncidentInmate

Requirements: FR-010, FR-012

Purpose: link incident to each inmate involved

Attributes: incident_id, inmate_id (composite primary key), role_in_incident (perpetrator/witness/victim) **Relationships:**

- Belongs to one incident and one inmate

Visitors**18. Visitor**

Requirements: FR-016

Purpose: A person's identity record.

Attributes: visitor_id, full_name, id_number, phone, relationship_to_inmate

Relationships:

- Has many Visits.

19. Visit

Requirements: FR-016 to FR-018 **Purpose:**

A single visit event.

Attributes: visit_id, visitor_id, inmate_id, visit_datetime, duration, logged_by

Relationships:

- Belongs to one Visitor and one Inmate.
- Logged by one User.

Rehabilitation

20. RehabProgram

Requirements: FR-022

Purpose: A program or activity, such as vocational training or counselling. **Attributes:** program_id, name, description, instructor, schedule, status

Relationships:

- Has many Enrollments.

21. Enrollment

Requirements: FR-022, FR-024

Attributes: enrollment_id, inmate_id, program_id, enrollment_date, status

Relationships:

- Belongs to one Inmate and one Program.
- Has many AttendanceRecords.

22. AttendanceRecord

Requirements: FR-023

Purpose: Per-session participation.

Attributes: attendance_id, enrollment_id, session_date, present (Y/N), remarks, recorded_by

Relationships:

- Belongs to one Enrollment.

System Support

23. AuditLog

Requirements: FR-025, FR-026

Purpose: A record of every create, update and delete.

Attributes: log_id, user_id, action_type, entity_name, record_id, old/new values, timestamp

Relationships:

- Belongs to one User.

24. Notification

Requirements: FR-027, FR-028

Purpose: Alerts for sentence completion and new incidents.

Attributes: notification_id, recipient_user_id, type, message, related_record_id, created_at, is_read

Relationships:

- Sent to one User.
- References a Sentence/Incident/health issues.

PRIMARY KEYS AND FOREIGN KEYS

The table below lists the primary key and foreign keys of each entity identified in the Prison Management System.

Entity	Primary Key	Foreign Key(s)	References
User	user_id	role_id	Role
Role	role_id	-	-
Inmate	inmate_id	-	-
Cell	cell_id	-	-
CellAssignment	assignment_id	inmate_id cell_id assigned_by	Inmate Cell User
Case	case_id	inmate_id recorded_by	Inmate User
Crime	crime_id	-	-
CaseCrime	case_id + crime_id (composite)	case_id crime_id	Case Crime
Sentence	sentence_id	case_id inmate_id	Case Inmate
Incident	incident_id	reported_by	User
IncidentInmate	incident_id + inmate_id (composite)	incident_id inmate_id	Incident Inmate
Entity	Primary Key	Foreign Key(s)	References
Visitor	visitor_id		-
Visit	visit_id	visitor_id inmate_id logged_by	Visitor Inmate User
RehabProgram	program_id	-	-

Enrollment	enrollment_id	inmate_id program_id	Inmate RehabProgram
AttendanceRecord	attendance_id	enrollment_id recorded_by	Enrollment User
AuditLog	log_id	user_id	User
Notification	notification_id	recipient_user_id	User
HealthRecord	health_record_id	inmate_id created_by	Inmate User
MedicalCondition	condition_id	health_record_id recorded_by	HealthRecord User
Consultation	consultation_id	health_record_id consulted_by	HealthRecord User
Medication	medication_id	consultation_id prescribed_by	Consultation User
MedicationAdministration	administration_id	medication_id administered_by	Medication User
MedicalAppointment	Appointment_id	health_record_id scheduled_by	HealthRecord User

Notes

1. CaseCrime and IncidentInmate have composite primary keys made up of their two foreign keys.